

TYUKHTET, Russian Federation

WMO: 294560

Lat: 56.536N

Lon: 89.317E

Elev: 673

StdP: 14.34

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.9	-30.9	-43.7	0.4	-38.7	-35.6	0.7	-30.5	18.9	16.6	16.9	12.7	1.6	230	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	83.9	66.0	80.7	64.4	77.6	63.3	69.1	78.5	67.1	76.8	65.3	74.7	5.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.9	74.3	63.6	90.1	71.7	61.7	84.1	69.9	33.8	78.3	32.1	77.3	30.6	75.0	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	12.8	11.4	DB	-41.3	89.4	8.9	3.1	-47.8	91.6	-53.0	93.4	-58.0	95.1	-64.5	97.4	(4)
(5)			WB	-41.0	71.1	7.5	3.9	-46.4	73.9	-50.7	76.2	-54.9	78.3	-60.3	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.1	-0.6	8.7	21.9	37.5	49.2	61.5	65.2	61.0	47.4	33.8	13.8	8.1		
	DBStd	25.09	16.92	13.64	11.86	8.82	10.07	8.18	5.98	6.65	8.44	9.75	15.76	15.81		
	HDD50	7187	1567	1156	872	388	145	13	0	5	148	507	1087	1299		
	HDD65	11449	2032	1576	1337	827	497	155	71	158	528	967	1537	1764		
	CDD50	1385	0	0	0	11	120	360	472	347	71	4	0	0		
	CDD65	172	0	0	0	0	7	51	78	35	1	0	0	0		
	CDH74	1610	0	0	0	3	95	583	632	276	21	0	0	0		
	CDH80	347	0	0	0	0	16	142	140	48	1	0	0	0		
	WSAvg	5.0	5.1	5.1	5.6	5.9	5.7	4.3	3.9	4.0	4.5	5.2	5.4	5.6		
	PrecAvg	20.9	0.9	0.8	0.9	1.2	1.7	2.6	2.4	2.9	2.0	2.1	1.9	1.5		
	PrecMax	26.0	1.6	1.5	2.4	2.0	4.2	4.6	4.4	6.5	3.5	3.6	2.8	2.7		
(15) Precipitation	PrecMin	16.2	0.1	0.3	0.2	0.3	0.6	0.6	0.8	1.2	0.2	0.8	1.0	0.5		
	PrecStd	2.4	0.4	0.3	0.5	0.4	0.9	0.9	0.9	1.2	0.9	0.6	0.5	0.6		
	DB	34.5	34.7	47.5	69.1	82.7	88.1	87.7	84.5	77.1	62.4	46.3	35.2			
	MCWB	30.8	30.1	38.3	51.2	61.5	65.8	69.8	67.5	60.7	49.2	41.9	32.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.0	31.6	43.5	62.1	76.5	84.2	83.7	80.5	70.3	55.2	40.4	32.3			
	2% MCWB	26.6	27.7	35.9	48.1	58.1	64.4	67.5	67.1	58.1	46.4	36.7	30.1			
	5% DB	25.2	29.4	40.6	56.8	72.0	80.3	80.4	76.8	65.9	50.7	36.7	29.2			
	5% MCWB	23.3	26.8	34.0	45.3	55.8	62.8	65.8	65.0	55.6	45.3	34.0	26.9			
	10% DB	22.0	26.8	37.7	52.3	67.2	76.8	77.3	73.3	61.1	46.7	33.2	26.1			
	10% MCWB	20.3	24.4	32.4	42.6	53.8	61.3	64.5	62.7	53.0	42.3	31.4	24.1			
	0.4% WB	31.2	31.2	39.6	52.9	63.5	69.8	72.8	71.3	61.1	51.1	42.2	32.9			
	0.4% MCDB	34.2	33.7	45.0	67.2	79.4	80.7	83.8	80.5	72.4	56.7	45.8	34.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	27.0	28.7	37.0	49.2	59.8	66.6	69.9	68.2	58.8	48.3	37.9	30.1			
	2% MCDB	28.8	30.8	42.3	58.7	72.1	79.8	79.2	77.1	70.0	54.4	39.5	32.0			
	5% WB	23.4	26.8	34.8	46.4	57.4	64.8	67.9	65.8	56.5	45.3	34.0	27.3			
	5% MCDB	25.1	29.7	39.2	56.3	69.6	77.0	77.1	74.5	64.3	50.3	36.5	29.2			
	10% WB	20.1	24.3	32.6	42.9	54.8	63.1	66.2	63.8	54.2	42.4	31.5	24.0			
	10% MCDB	22.0	26.8	37.2	51.3	65.7	73.1	75.0	71.4	60.2	46.7	33.2	25.8			
	MDBR	13.0	14.2	16.4	16.1	18.2	19.0	17.5	17.1	15.3	11.1	10.8	11.4			
	5% DB	12.3	11.9	18.1	22.8	25.1	25.9	22.8	22.2	24.5	17.5	10.4	10.9			
(36) Mean Daily Temperature Range	MCWBR	11.3	9.9	12.5	12.7	12.6	10.2	9.8	10.5	13.1	11.1	8.6	9.7			
	MCDBR	12.5	12.0	14.4	21.4	21.9	21.7	19.4	19.3	22.0	15.8	9.6	10.5			
	MCWBR	11.6	10.6	10.6	13.1	11.7	9.7	9.7	10.3	12.3	10.5	8.3	9.9			
	MCDBR	11.6	10.6	10.6	13.1	11.7	9.7	9.7	10.3	12.3	10.5	8.3	9.9			
(40) Clear-Sky Solar Irradiance	taub	0.290	0.322	0.370	0.398	0.408	0.407	0.433	0.409	0.372	0.351	0.319	0.290			
	taud	1.986	1.966	1.907	2.064	2.202	2.294	2.220	2.270	2.321	2.218	2.024	1.935			
	Ebn at Noon	174	219	239	256	263	265	254	252	245	212	159	134			
	Edh at Noon	21	32	45	45	42	38	41	36	30	25	20	17			
(44) All-Sky Solar Radiation	RadAvg	236	540	959	1345	1565	1865	1698	1342	865	439	211	138			
	RadStd	30	49	74	107	173	173	118	105	113	40	30	20			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page